

In the following exercises, all coordinates and components are given with respect to an orthonormal frame $\mathcal{K} = (O, \mathcal{B})$. For the 2-dimensional cases, assume $\mathcal{B} = (\mathbf{i}, \mathbf{j})$, and for the 3-dimensional cases, assume $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$.

3.1. Let $\mathbf{m}$ and $\mathbf{n}$ be two unit vectors such that $\angle(\mathbf{m}, \mathbf{n}) = 120^\circ$. Determine the angle between the vectors $\mathbf{a} = 2\mathbf{m} + 4\mathbf{n}$ and $\mathbf{b} = \mathbf{m} - \mathbf{n}$.

3.2. Consider the points $A(1, 2)$, $B(3, -2)$, and $C(5, 6)$. For the triangle ABC :

- a) determine the circumcenter,
- b) determine the orthocenter.

3.3. Consider the points $A(5, 10)$, $B(19, 10)$, $C(10, 22)$. Determine the incenter of the triangle ABC .

3.4. Determine a Cartesian equation of the plane π if $A(1, -1, 3)$ is the orthogonal projection of the origin onto π .

3.5. Let $A(1, 2, -7)$, $B(2, 2, -7)$, and $C(3, 4, -5)$ be vertices of a triangle. Determine equations for the angle bisector of the angle $\angle CAB$.

3.6. Determine the planes which pass through $P(0, 2, 0)$ and $Q(-1, 0, 0)$ and which form an angle of 60° with the z -axis.

3.7 (Law of Cosines). Let ABC be a triangle with $|AB| = c$, $|BC| = a$, and $|CA| = b$. Show that

$$\langle \overrightarrow{AB}, \overrightarrow{AC} \rangle = \frac{1}{2}(b^2 + c^2 - a^2)$$

and deduce the law of cosines in a triangle.

3.8 (Heron's Formula). Let ABC be a triangle with $|AB| = c$, $|BC| = a$, $|CA| = b$, and let p denote the semiperimeter $\frac{1}{2}(a + b + c)$. Show that

$$\langle \overrightarrow{AB}, \overrightarrow{AC} \rangle + bc = 2p(p - a) \quad \text{and} \quad \langle \overrightarrow{AB}, \overrightarrow{AC} \rangle - bc = -2(p - b)(p - c).$$

Deduce that the area of the triangle ABC is given by

$$\text{Area}(ABC) = \sqrt{p(p - a)(p - b)(p - c)}.$$